

On the selection of active slip systems in crystal plasticity

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Abstract

The capabilities of existing rate-independent and rate-dependent constitutive models to select the active slip systems at the corners of non-smooth theories play a crucial role in predicting localisation phenomena. Even though the description of crystal plasticity within the context of modern continuum mechanics goes back to the early 1960s, there is no universally accepted solution as to how to identify a unique set of active slip systems. Furthermore, some recently proposed integration schemes have neither been compared with other methods nor tested under complex multiaxial stress conditions thus rendering a direct assessment difficult. In this work, the predictive capabilities of existing crystal plasticity and visco-plasticity formulations and algorithms when subjected to complex multiaxial loading paths are investigated, and their relative accuracies established. In order to compare consistently the performance of different models, a generic thermodynamics-based crystallographic framework, which incorporates current formulations as special cases, is proposed. Several two-dimensional boundary value problems for elasto-plastic and elasto-viscoplastic FCC crystals are selected as benchmark cases. The effects of multiaxial loading paths, latent hardening, and dissipated energy on the selection of active slip systems at sharp yield surface corners are investigated. The differences in the predicted behaviour were found to be associated with both the particular form of the single crystal formulations and the algorithms used in their numerical implementations. Experimental data

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obtained under multiaxial loading conditions will be required to judge the relative accuracy of each single crystal formulation.

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1. Introduction

Constitutive models possessing non-smooth yield or potential surfaces are widely used in applications such as soil and rock mechanics and crystal plasticity. Crystal plasticity models in particular rely on the understanding that plastic deformation in crystals at low homologous temperatures occurs by crystallographic slip following Schmidt's law. The typical distortion experienced by an FCC single crystal yield surface under an imposed biaxial strain path (predicted stress paths given by $a \rightarrow f$) is shown in Fig. 1(a) (Busso and Cailletaud, 2003). The shape of the yield surface

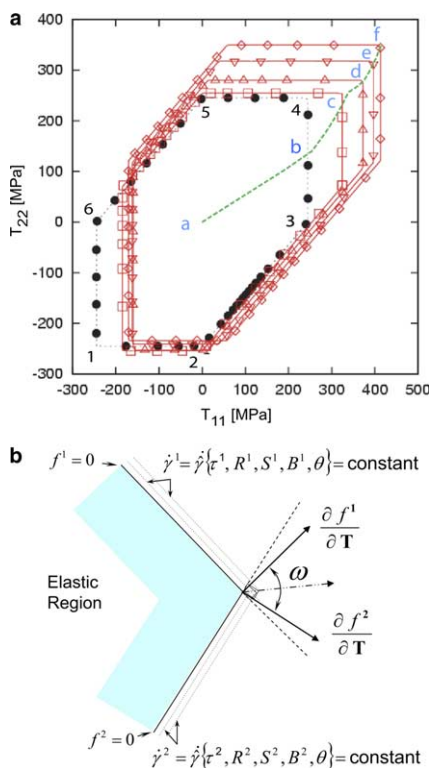


Fig. 1. (a) Distortion of a typical FCC single crystal yield/potential surface under biaxial loading ($E_{11}/E_{22} = E_{[1\ 0\ 0]}/E_{[0\ 1\ 0]} = 5$, $\dot{E}_{11} = 0.01\text{ s}^{-1}$) and (b) geometric interpretation of a singular point in non-smooth yield/potential surfaces.

depends both on the type of loading and on the slip systems activated during plastic flow, particularly when the current stress state is found at the corner of the yield surface (Points d , e and f in Fig. 1(a)).

The geometric interpretation of a singular point at the intersection of two slip systems' yield surfaces ($f^1 = 0$ and $f^2 = 0$) is given in Fig. 1(b). Here the directions of the strain increment vectors are confined to a narrow region between the normals to the adjacent sides, i.e. the vectors $\partial f^i / \partial \mathbf{T}$ (for $i = 1, 2$), defined by the angle ω in Fig. 1(b). However, in rate-independent crystallographic models, the direction of plastic flow within corner regions such as this is not uniquely defined.

The selection of the active slip systems is known to lead to an ill-conditioned problem for certain hardening descriptions, some of which may not be physically meaningful. In the original work of Taylor (1938), it was proposed that the slip systems which should be operative are those for which the energy dissipated is the least (minimum internal work principle). Taylor's minimisation of internal work is equivalent to what Bishop and Hill (1951) later showed to be a maximisation of the plastic work but only when the hardening in all systems is equal. However, a generalisation of Bishop and Hill's idea requires that the identification of the active systems be such so that they maximise the plastic dissipation. In Hill's work (1966), it is shown that the issue of uniqueness of slips in rate-independent crystal plasticity is intimately linked to the slip systems latent hardening behaviour. There, it can be seen that if the slip system hardening matrix is positive definite, then uniqueness would be satisfied. For instance, the assumption of simple Taylor hardening does not lead to a positive definite matrix. Hardening models such as that by Bassani (1994) have been proposed which assure the positive definiteness of the hardening matrix. In addition, the way in which rate-independent theories are implemented numerically also plays a crucial role in the way the active slip systems are selected and the slip rates determined (Busso and Cailletaud, 2003; Schmidt-Baldassari, 2003). Nevertheless, the emphasis of the work presented here will be on assessing the capabilities of existing single crystal models and algorithms to predict complex multiaxial loading conditions rather than on a detailed investigation of the underlying physics of the hardening behaviour described by each single crystal formulation.

The description of crystal plasticity within the context of modern continuum mechanics dates back to the work of Mandel (1965) and Hill (1966), and the further developments by Hill and Rice (1972), Asaro and Rice (1977), Asaro (1983) and Peirce et al. (1983). More recently, complex numerical algorithms have been proposed to identify the active sets of slip systems in rate-independent crystallographic models, e.g. Anand and Kothari (1996), Simo and Hughes (1998), McGinty (2001), Schmidt-Baldassari (2003), amongst others. However, the relative importance of the constitutive formulation and numerical algorithms with respect to the selection of active slip systems is not clear. Furthermore, recently proposed integration schemes have neither been compared with other schemes nor tested under complex multiaxial stress conditions thus rendering a direct assessment difficult.

One way of overcoming the non-uniqueness problem associated with the identification of the active slip systems is to work within a rate-dependent framework. There, slip is no longer described by a rate-independent yield surface (e.g. full lines

in Fig. 1(b)) but is instead controlled by a rate-dependent potential surface (e.g. dotted lines in Fig. 1(b)). Most importantly, due to the smooth nature of the viscous potential function at the intersection between two slip system surfaces (i.e. region defined by ω in Fig. 1(b)), the direction of the strain increment vectors can be readily obtained following the normality rule. It is nevertheless worth pointing out that rate-dependent constitutive formulations used in the quasi-rate independent regime generally exhibit a very stiff behaviour which can lead to unacceptably large computational times.

In previous work (Busso and Cailletaud, 2003), the capabilities of typical rate-dependent crystal plasticity formulations to predict complex deformation behaviour were discussed. Here, that work is extended to rate-independent plasticity formulations under similar complex loading conditions. An investigation into the predictive capabilities of existing rate-dependent and rate-independent models to predict complex multiaxial loading paths is conducted, and their relative performances compared. A general thermodynamics-based crystallographic framework is proposed to compare consistently the performance of the different crystallographic models. For simplicity, only small strain and isothermal conditions will be considered.

2. General thermodynamics single crystal framework

2.1. Generic slip system flow and evolutionary equations

A general expression for the flow stress or critical resolved shear stress in a generic slip system α is given by

$$\tau^\alpha = \pm \hat{f}^\alpha \{ \dot{\gamma}^\alpha, R^\alpha, \theta \} \pm c_s S^\alpha + B^\alpha, \quad (1)$$

where $\dot{\gamma}^\alpha$ is the slip rate, $\hat{f}^\alpha$ is a function which represents viscous effects, θ is the absolute temperature, c_s is a switching parameter ($= 0$ or 1), and R^α , S^α and B^α are internal slip system variables which represent multiplicative and additive slip resistances associated with non-directional hardening mechanisms, and internal stresses associated with kinematic hardening processes, respectively. Note that the distinction between a multiplicative (R^α) and an additive (S^α) slip resistance in the first and second right hand-side terms of Eq. (1) is motivated by the additive and multiplicative use of non-directional hardening variables rather than on mechanistic considerations. Such need should become more apparent when the different single crystal models are derived from this generic formulation. By expressing the slip system flow stress in the way shown in Eq. (1), the contributions from viscous effects, and directional and non-directional hardening mechanisms can be clearly identified.

In a rate-dependent crystallographic formulation, the slip rate or *flow rule* can be conveniently found by inverting Eq. (1),

$$\dot{\gamma}^\alpha = \hat{\gamma}^\alpha \{ \tau^\alpha, R^\alpha, S^\alpha, B^\alpha, \theta \}. \quad (2)$$

A rate-independent crystallographic formulation, in contrast, can only be expressed in terms of an additive slip resistance, i.e. S^α in Eq. (1) and with $c_s = 1$. Here, the actual magnitudes of the slip rates for the active slip systems are found by setting

the viscous contribution in Eq. (1) to zero, that is $f^\alpha = 0$. Thus the yield condition is obtained directly from Eq. (1) as

$$|\tau^\alpha - B^\alpha| = S^\alpha. \quad (3)$$

Those slip systems where the condition given by Eq. (3) is satisfied correspond to the facets of Fig. 1's yield surface. Furthermore, the consistency condition stipulate that during plastic deformation, the condition which must hold for all the active systems is

$$|\dot{\tau}^\alpha - \dot{B}^\alpha| = \dot{S}^\alpha. \quad (4)$$

Eq. (4) yields a system of equations which is restricted to the active slip systems which simultaneously satisfy,

$$f^\alpha = 0 \quad \text{and} \quad \dot{f}^\alpha = 0. \quad (5)$$

The resolved shear stress is defined as the work conjugate to the slip rate,

$$\tau^\alpha = \mathbf{T} : \mathbf{P}^\alpha \quad \text{with} \quad \mathbf{P}^\alpha = \frac{1}{2}(\mathbf{n}^\alpha \otimes \mathbf{m}^\alpha + \mathbf{m}^\alpha \otimes \mathbf{n}^\alpha), \quad (6)$$

where $\mathbf{T}$ is the stress tensor, $\mathbf{m}^\alpha$ and $\mathbf{n}^\alpha$ are the slip system unit vectors defining the slip direction and the slip plane normal, respectively, and $\otimes$ denotes the tensorial product.

To complete the crystallographic formulation for a generic slip system, the evolutionary behaviour of the internal slip system variables is required to describe the distortion of the yield or potential surfaces during deformation. Let the time rate of change of each internal slip system variable be functionally defined as,

$$\dot{R}^\alpha = \hat{R}^\alpha \{ \dot{\gamma}^\alpha, R^\alpha, R^\beta, S^\alpha, B^\alpha, \theta \}, \quad (7)$$

$$\dot{S}^\alpha = \hat{S}^\alpha \{ \dot{\gamma}^\alpha, R^\alpha, S^\alpha, S^\beta, B^\alpha, \theta \}, \quad (8)$$

$$\dot{B}^\alpha = \hat{B}^\alpha \{ \dot{\gamma}^\alpha, R^\alpha, S^\alpha, B^\alpha, \theta \}. \quad (9)$$

Note that the dependencies of Eq. (7) on R^β and of Eq. (8) on S^β are introduced to account for cross-hardening effects in the evolutionary behaviour of the slip resistances.

In a thermodynamics framework, Eqs. (7)–(9), are derived from those of their corresponding state variables, as it will be shown next. It should be noted that the latter, in contrast to internal microstructural variables, are by definition unaffected by instantaneous changes in temperature.

2.2. Associated crystallographic formulation

In a generic associated crystallographic formulation expressed within a thermodynamics framework, the slip system hardening variables R^α , S^α and B^α can be explicitly expressed in terms of their corresponding thermodynamical state variables, here denoted r^α , s^α and b^α , respectively. The inelastic part of the free energy function can be expressed in terms of generic functions of r^α , s^α and b^α as follows:

$$\rho_0 \psi = \sum_{\alpha} (\psi_r^{\alpha} + \psi_s^{\alpha} + \psi_b^{\alpha}), \quad (10)$$

where the sum on (α) extends over all potentially active slip systems, ρ_0 is the material density, and $\psi_r^{\alpha} = \hat{\psi}_r^{\alpha}\{r^{\alpha}\}$, $\psi_s^{\alpha} = \hat{\psi}_s^{\alpha}\{s^{\alpha}\}$ and $\psi_b^{\alpha} = \hat{\psi}_b^{\alpha}\{b^{\alpha}\}$ are explicit functions of the state variables. Note that, for simplicity, no coupling between the state variables has been considered in these functions.

The relations between the slip system internal variables and their corresponding state variables can be found from Eq. (10), since, by definition, they can be linked through the free energy derivatives:

$$R^{\alpha} = \rho_0 \frac{\partial \psi}{\partial r^{\alpha}} = \frac{\partial \psi_r^{\alpha}}{\partial r^{\alpha}}, \quad (11)$$

$$S^{\alpha} = \rho_0 \frac{\partial \psi}{\partial s^{\alpha}} = \frac{\partial \psi_s^{\alpha}}{\partial s^{\alpha}}, \quad (12)$$

$$B^{\alpha} = \rho_0 \frac{\partial \psi}{\partial b^{\alpha}} = \frac{\partial \psi_b^{\alpha}}{\partial b^{\alpha}}. \quad (13)$$

From the linearity of the stress rate and the elastic strain rate tensors in the Clausius–Duhem inequality, the stress tensor can be expressed in terms of the material density and the derivative of the total free energy with respect to the elastic strain tensor.

Furthermore, the thermodynamical formulation of a rate-dependent crystallographic model for infinitesimal strains requires a generic viscoplastic potential,

$$\Omega = \sum_{\alpha} \hat{\Omega}^{\alpha}\{f^{\alpha}\} \quad \text{with } f^{\alpha} = \hat{f}^{\alpha}\{\tau^{\alpha}, R^{\alpha}, S^{\alpha}, B^{\alpha}, \theta\}, \quad (14)$$

so that the time rate of change of the inelastic strain tensor linked to the motion of dislocations in N slip systems is

$$\dot{\epsilon}^p = \frac{\partial \Omega}{\partial \mathbf{T}} = \sum_{\alpha} \frac{\partial \Omega}{\partial f^{\alpha}} \frac{\partial f^{\alpha}}{\partial \mathbf{T}} = \sum_{\alpha} \dot{\gamma}^{\alpha} \mathbf{P}^{\alpha}. \quad (15)$$

The summation in Eq. (15) extends over all active slip systems.

It is worth recalling that the common framework of all the present crystal plasticity models is the thermodynamically consistent one defined by Rice (1971), where the free energy appears as a thermodynamical potential linking the state variables with the hardening variables. When a second dissipation potential is used to derive the viscoplastic flow and to characterize the elastic domain, then the constitutive formulation is said to be *associated*. However, these models cannot be considered to belong to the framework of generalised standard plasticity (Germain et al., 1983), since the evolutionary behaviour of the internal slip system variables is not constrained to be derived from the same dissipation potential given in Eq. (14), enabling the evolutionary equations to be formulated directly from the governing hardening-recovery mechanisms. Thus the framework presented here is suitable for associated non-generalised formulations.

2.3. Dissipated energy

It should be emphasised that the main purpose behind the formulation of the general thermodynamics-based crystallographic framework presented here is to enable the dissipated energies predicted by different single crystal models to be compared consistently. Note also that the reformulation of an existing single crystal model within the proposed thermodynamic framework does yield exactly the same stress–strain response as that obtained using the constitutive equations in their original form.

The intrinsic dissipated energy for an associated crystallographic model is given by the difference between the plastic work and the internal dissipation due to non-directional and directional (i.e. kinematic) hardening mechanisms and, for rate-dependent formulations, viscosity. Thus for a rate-dependent framework and for formulations with additive slip resistances, the dissipation is expressed as

$$\phi = \mathbf{T} : \dot{\mathbf{e}}^p - \sum_{\alpha} \left(\sum_{\beta} \rho_0 \frac{\partial \psi}{\partial \psi_s^{\beta}} \frac{\partial \psi_s^{\beta}}{\partial s^{\alpha}} \right) \dot{s}^{\alpha} - \sum_{\alpha} \left(\sum_{\beta} \rho_0 \frac{\partial \psi}{\partial \psi_b^{\beta}} \frac{\partial \psi_b^{\beta}}{\partial b^{\alpha}} \right) \dot{b}^{\alpha}, \quad (16)$$

and for those with a multiplicative slip resistance as,

$$\phi = \mathbf{T} : \dot{\mathbf{e}}^p - \sum_{\alpha} \left(\sum_{\beta} \rho_0 \frac{\partial \psi}{\partial \psi_r^{\beta}} \frac{\partial \psi_r^{\beta}}{\partial r^{\alpha}} \right) \dot{r}^{\alpha} - \sum_{\alpha} \left(\sum_{\beta} \rho_0 \frac{\partial \psi}{\partial \psi_b^{\beta}} \frac{\partial \psi_b^{\beta}}{\partial b^{\alpha}} \right) \dot{b}^{\alpha}. \quad (17)$$

In rate-independent crystallographic formulations, the viscous function f^{α} in Eq. (1) is equal to zero and thus the multiplicative slip resistance R^{α} ceases to have any meaning. The dissipation in such cases is then given by Eq. (16) since non-directional hardening (or softening) processes can only accommodate an additive type of slip resistance, see Eq. (4).

From the partial derivatives of Eq. (10), with Eqs. (12) and (13), and the definition of the plastic work, $\mathbf{T} : \dot{\mathbf{e}}^p = \sum_{\alpha} \dot{\gamma}^{\alpha} \tau^{\alpha}$, the dissipation given by Eq. (16) becomes,

$$\phi = \mathbf{T} : \dot{\mathbf{e}}^p - \sum_{\alpha} (S^{\alpha} \dot{s}^{\alpha} + B^{\alpha} \dot{b}^{\alpha}), \quad (18)$$

$$= \sum_{\alpha} (\dot{\gamma}^{\alpha} \tau^{\alpha} - S^{\alpha} \dot{s}^{\alpha} - B^{\alpha} \dot{b}^{\alpha}). \quad (19)$$

Note that, generally, a given crystal plasticity formulation should automatically yield a positive dissipation in Eq. (19) for materials exhibiting strain hardening.

In the next section, the deformation behaviour predicted by typical single crystal formulations, including the particular form of the equations required to compute Eq. (19), will be presented.

3. Single crystal formulations

In this work, the relative predictive capabilities of four distinct rate-dependent and two rate-independent crystallographic formulations will be investigated. The

rate-dependent single crystal models are those of Meric et al. (1991a,b); Cailletaud (1987), Busso (1990) (see also Busso and McClintock, 1996; Cheong and Busso, 2004), Anand and Kothari (1996), and McGinty and McDowell (McGinty, 2001). One of the main differences between these models is the particular form of their flow rules. The Meric and Cailletaud and the Busso models, henceforth to be referred to as Models GC and EB, respectively, incorporate an *additive* slip resistance (i.e. S^α in Eq. (1)) in their flow rules. In contrast, those by Anand and Kotari and by McGinty and McDowell, to be referred to as Models LA, and DM, respectively, incorporate a *multiplicative* slip resistance (i.e. R^α in Eq. (1)). The implications of this difference will be discussed later in the text. The rate-independent single crystal models available for comparison where those of Meric et al. (1991a,b) and Anand and Kothari (1996). Henceforth, these rate-independent models will also be referred to as Models GC and LA, respectively.

Tables 1 and 2 summarize the relevant flow and evolutionary equations for each single crystal model, expressed within the thermodynamics framework presented in Section 2. Note that, to compute the dissipated energy predicted by each model, the relations between slip system variables ($R^\alpha, S^\alpha, B^\alpha$) and their corresponding state variables ($r^\alpha, s^\alpha, b^\alpha$) must be given. These relations can be found in the last three rows of Table 1. The evolutionary equations for the slip resistances of the GC and LA models are assumed to be the same for the rate-independent and rate-dependent versions of those models. It should also be noted that the rate-independent GC

Table 1
Functions for the different single crystal models

Function	GC	EB	LA	DM
f^α	$ \tau^\alpha - B^\alpha - S^\alpha - S_0$	$ \tau^\alpha - B^\alpha - S^\alpha$	$ \tau^\alpha - B^\alpha $	$ \tau^\alpha - B^\alpha $
$ \dot{\gamma}^\alpha $	$\dot{\gamma}_0 \langle \frac{f^\alpha}{\tau_0} \rangle^n$	$\dot{\gamma}_0 \exp\{-\frac{f_0}{k\theta} \langle 1 - \langle \frac{f^\alpha}{\tau_0} \rangle^p \rangle^q\}$	$\dot{\gamma}_0 \langle \frac{f^\alpha}{R^\alpha} \rangle^n$	$\dot{\gamma}_0 \langle \frac{f^\alpha}{R^\alpha} \rangle^n$
$\psi_{\frac{1}{2}}^\alpha \{r^\alpha\}$	—	—	$\frac{1}{2} h_s (r^\alpha)^2$	$\frac{1}{2} h_s (r^\alpha)^2$
$\psi_{\frac{1}{2}}^\alpha \{s^\alpha\}$	$\frac{1}{2} h_s s^\alpha \sum_\beta h^{2\beta} s^\beta$	$\frac{1}{2} h_s \alpha (s^\alpha)^2$	—	—
$\psi_b^\alpha \{b^\alpha\}$	$\frac{1}{2} h_b (b^\alpha)^2$	$\frac{1}{2} h_b (b^\alpha)^2$	$\frac{1}{2} h_b (b^\alpha)^2$	$\frac{1}{2} h_b (b^\alpha)^2$
R^α	—	—	$h_s r^\alpha$	$h_s r^\alpha$
S^α	$h_s \sum_\beta h^{2\beta} s^\beta$	$h_s s^\alpha$	—	—
B^α	$h_B b^\alpha$	$h_B b^\alpha$	$h_B b^\alpha$	$h_B b^\alpha$

Table 2
Evolutionary equations for the different single crystal models

Model	Equation
$\dot{S}_{GC}^\alpha =$	$\sum_\beta h^{2\beta} [h_s - d_s s^\beta] \dot{\gamma}^\beta $
$\dot{S}_{EB}^\alpha =$	$\sum_\beta h^{2\beta} (h_s - d_s [S^\beta - S_0]) \dot{\gamma}^\beta $
$\dot{R}_{LA}^\alpha =$	$\sum_\beta h_s [q_I + (1 - q_I) \delta^{2\beta}] \{1 - \frac{R^\beta}{R^\alpha}\}^a \dot{\gamma}^\beta $
$\dot{R}_{DM}^\alpha =$	$\sum_\beta q_I h_s \dot{\gamma}^\beta - (q_I - 1) h_s \dot{\gamma}^\alpha - d_s R^\alpha [\sum_\beta q_I \dot{\gamma}^\beta - (q_I - 1) \dot{\gamma}^\alpha]$
$\dot{B}^\alpha =$	$[h_B \text{sign}(\dot{\gamma}^\alpha) - d_B B^\alpha] \dot{\gamma}^\alpha $

model can be obtained directly from its rate-dependent version by setting the viscous contribution to the flow stress to zero ($f^v = 0$ in Eq. (1)), as previously discussed. All models have been numerically implemented into the finite element method using the implicit numerical integration schemes detailed in the corresponding references. No details of the integration schemes will be given here for brevity.

The monotonic stress–strain behaviour of a strain-hardening FCC material in the rate-independent regime, i.e. at $\theta = 25^\circ\text{C}$, was considered. Furthermore, twelve (i.e. $N = 12$) $\{111\}\langle 110 \rangle$ octahedral systems were assumed to be potentially active. The calibration of the rate-dependent model parameters was done under uniaxial $\langle 100 \rangle$ loading in the quasi-rate-independent regime assuming Taylor hardening. The calibration was done so that the rate-dependent response of the material for strain rates ranging from 10^{-1} – 10^{-3} s^{-1} was acceptably small. Similarly, calibration of the rate-independent GC and LA hardening parameters was done under $\langle 100 \rangle$ uniaxial loading with Taylor hardening. For both types of models, no kinematic hardening was assumed for simplicity (i.e. $\dot{B}^x = 0$). The list of the calibrated coefficients for each model is given in Table 3, and the corresponding predicted uniaxial responses obtained from the rate-dependent models for applied strain rates of 10^{-3} and 10^{-1} s^{-1} are shown in Fig. 2.

Table 3
Material coefficients used in the different single crystal models

Coefficient	Rate-dependent model parameters				Rate-independent model parameters		Units
	GC	EB	LA	DM	GC	LA	
C_{11}	250	250	250	250	250	250	GPa
C_{12}	200	200	200	200	200	200	GPa
C_{44}	100	100	100	100	100	100	GPa
$\dot{\gamma}_0$	1.0	150	7.3×10^{-4}	1.0	–	–	s^{-1}
n	2.0	–	0.01	0.01	–	–	–
$\hat{\tau}_0$	20	465	–	–	–	–	MPa
F_0	–	48.9	–	–	–	–	kJ mol
p	–	0.163	–	–	–	–	–
q	–	1.220	–	–	–	–	–
S_0	100	105	103	108	100	–	MPa
R_0	–	–	103	108	–	100	MPa
R^*	–	–	195	–	–	235	MPa
a	–	–	0.929	–	–	1.6	–
h_S	1008	1150	1824	2203	1092	1953	MPa
d_S	84672	12.1	–	10.5	9532	–	–
h_B	0.0	0.0	0.0	0.0	0.0	0.0	MPa
d_B	0.0	0.0	0.0	0.0	0.0	0.0	–
$h^{z\beta}, q_I$ (Self)	0	0	0	0	0	0	–
$h^{z\beta}, q_I$ (Taylor)	1	1	1	1	1	1	–

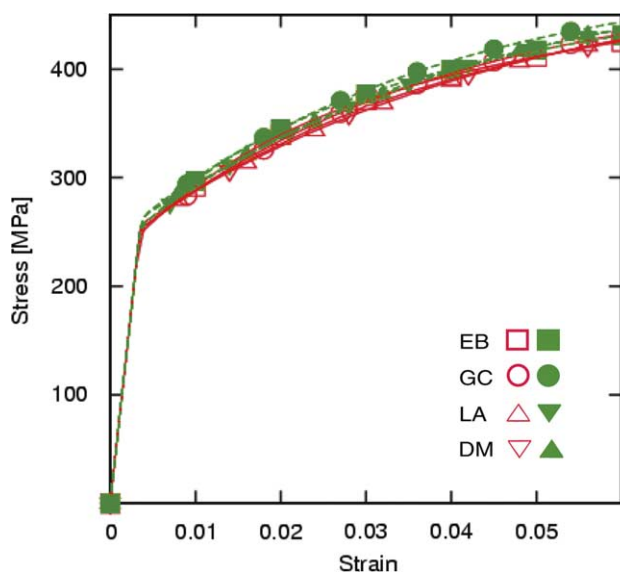


Fig. 2. Uniaxial rate-dependent model predictions for strain rates of 10^{-3} s^{-1} (open symbols), and 10^{-2} s^{-1} (filled symbols) at 25°C .

4. Results and discussions

4.1. Rate-dependent models

Small strain finite element calculations of complex biaxial loading cases were performed with the different single crystal models using special representative volume elements, which enable the shear and normal strains to be independently specified as imposed boundary conditions. Note that this will not be possible using standard displacement-based finite elements or large strain assumptions. The finite element calculations were performed with the code Zebulon (Besson et al., 1998). In all cases to be discussed below, the $[100]$ crystallographic axis is always parallel to the X_1 reference axis, and the applied $\dot{E}_{11}$ rate is 10^{-2} s^{-1} .

Fig. 3(a) shows the biaxial stress paths predicted by the rate-dependent formulations as the result of imposed E_{11}/E_{22} ratios of -0.95 , -5.00 and 5.33 for a material with Taylor hardening (i.e. $h^{z\beta} = q_l = 1.0$ in Table 3). The initial yield surface is also shown for reference. It can be seen that the different models predict similar plastic flow directions for each of the three different strain paths. To provide some insight into the individual slip systems which are activated, Fig. 3(b) shows the slip accumulated in the different octahedral systems predicted by the EB model in terms of the T_{11} stress for the $E_{11}/E_{22} = 5.33$ case shown in (a). It can be seen that, at approximately 250 MPa, a set of four primary $\{111\}\langle 110 \rangle$ slip systems are activated, and a secondary set of other four systems (corresponding to the horizontal yield surface in Fig. 3(a)) becomes active at approximately 310 MPa.

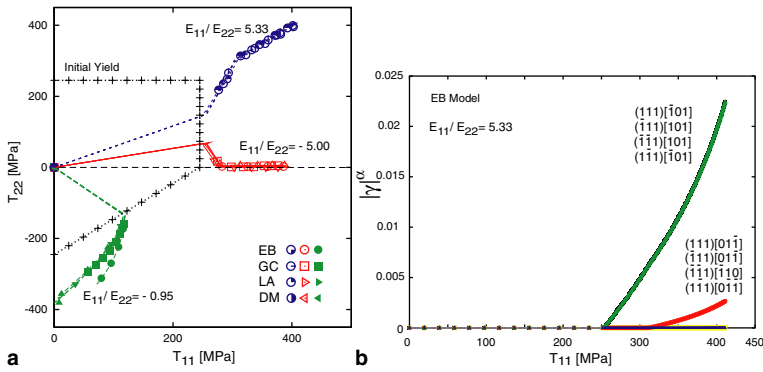


Fig. 3. T_{11} - T_{22} biaxial loading paths under different imposed E_{11}/E_{12} ratios (X_{11} parallel to $[100]$): (a) Comparison between the four **rate-dependent** models with only non-directional hardening ($\dot{R}^z, \dot{S}^z \neq 0, \dot{B}^z = 0$) and with Taylor latent hardening and (b) the corresponding accumulated slip in the octahedral systems predicted by the EB model vs. T_{11} for the $E_{11}/E_{22} = 5.33$ case shown in (a) ($\dot{\epsilon}_{11} = 10^{-2} \text{ s}^{-1}$, $\theta = 25^\circ \text{C}$).

To bring up more strongly the effect of latent hardening on slip system activation, biaxial normal-shear stress paths due to imposed E_{11}/E_{12} ratios of 0.02 and 0.40 were computed under Taylor and self-hardening (i.e. $h^{z\beta} = q_l = 0.0$, see Table 3) assumptions. The results from the four rate-dependent models are shown in Figs. 4(a) and (b), respectively. The initial yield surface and that of the relevant latent system are also shown. The most interesting aspects here are the contrasting predictions obtained from the EB and GC Models, on one hand, and the LA and DM, on the other, in particular for Taylor hardening. It is also worth noticing in Fig. 4(a) the sudden activation of the secondary latent system at approximately $T_{11} = 21$ and 35 MPa for the LA and the DM Models, respectively.

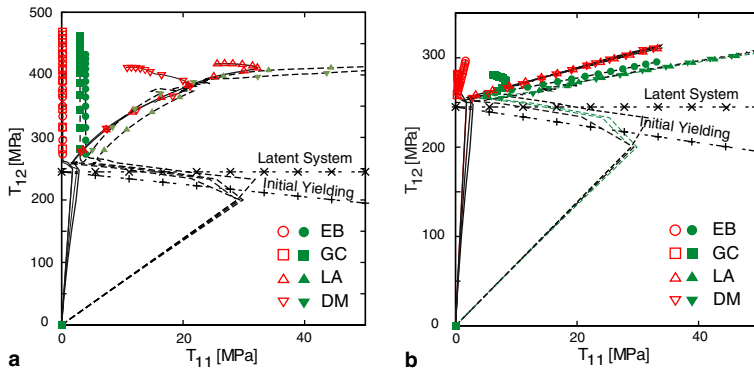


Fig. 4. Comparison between the **rate-dependent** model predictions of T_{11} - T_{12} biaxial stress paths under imposed E_{11}/E_{12} strain ratios of 0.02 (open symbols) and 0.40 (filled symbols) with (a) Taylor and (b) latent hardening ($\dot{\epsilon}_{11} = 10^{-2} \text{ s}^{-1}$, $\theta = 25^\circ \text{C}$).

The discrepancy between the two sets of models arises due to the fact that, in the DM and LA models, the updating of the crystal positions is actually part of the large strain formalism used in their numerical implementation, which is not the case for the EB and GC models. This updating does not play a role for the case of biaxial loading (11–22), but can have a significant effect in (11–12) biaxial loading case. To illustrate this effect, the MD model predictions shown in Fig. 4(a) are compared in Fig. 5 with a case where the rotations have been enforced to be zero (labelled as MD-NR in Fig. 5). It can be seen that the response of the latter is now much close to those of the GC and EB models.

Finally, Figs. 6(a)–(d) show the accumulated slip predicted by the four rate-dependent models for the $E_{11}/E_{12}=0.02$ case shown in Fig. 4(a) in terms of the T_{11} stress. Here, different lines correspond to different slip systems. As expected, the predicted accumulated slip in the different octahedral systems by the EB and GC models are starkly different from those of the LA and DM models. In the latter two cases, the inflection in the curves indicates the activation of the latent system.

It should be pointed out that the implicit numerical integration method used to update the incremental values of the stress and internal variables in each rate-dependent model strongly conditions the predictions obtained in Figs. 4 and 6. One of the most contrasting differences between these algorithms is the choice made for the roots of the relevant non-linear system of equations solved by the Newton-type iterative scheme at the local (i.e. material point) level. For instance, the numerical algorithm in the GC model (Besson et al., 2001) is inspired by Simo’s work (Simo and Hughes, 1998). There, a θ -method is used, in which the unknowns are the increments of the elastic strain tensor components and the incremental slip on each active

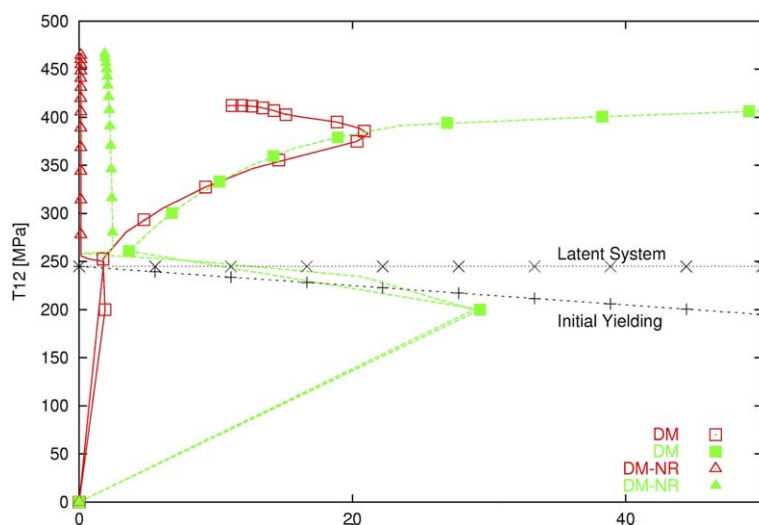


Fig. 5. Illustration of the role of the rotation for tension–shear loading: MD model compared with MD model without crystal update (MD-NR) for the case of Taylor hardening under imposed E_{11}/E_{12} strain ratios of 0.02 (open symbols) and 0.40 (filled symbols).

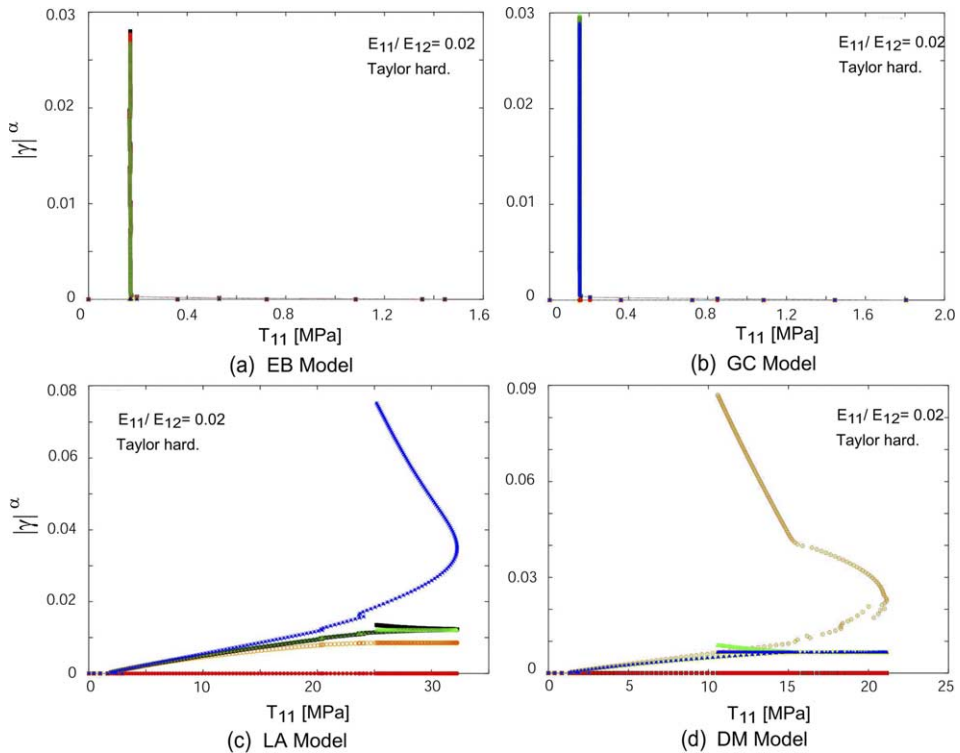


Fig. 6. Accumulated slip predicted by the four *rate-dependent* models for the $E_{11}/E_{12} = 0.02$ case shown in Fig. 4(a) (Taylor hardening) as a function of the $[1\ 0\ 0]$ stress component (T_{11}): (a) EB, (b) GC, (c) LA and (d) DM.

system. The local system is then solved by means of a Newton-type method. The classical numerical procedure proposed by [Simo and Hughes \(1998\)](#) is also used to determine the consistent tangent matrix. In contrast, the EB model (e.g. see [Meisnier et al., 2001](#)) relies on an implicit Euler-backward method and a Newton–Raphson scheme with a single level of iteration to solve for the incremental values of the stresses and slip system internal variables. The other two models follow a similar approach to the latter. In ongoing work, the relative effects of the constitutive formulations per se and the integration schemes used at the local level are being investigated.

4.2. Rate-independent models

Similar small strain finite element calculations to those performed with the rate-dependent models were carried out with the GC and LA rate-independent ones using the user-defined representative volume elements. All calculations were done assuming non-directional hardening only with a Taylor assumption (i.e. $\dot{B}^z = 0$, $h^{z\beta} = q_l = 1.0$).

Fig. 7 shows the biaxial T_{11} – T_{22} stress paths predicted by the rate-independent models as the result of imposed E_{11}/E_{22} ratios of -0.95 , -5.00 and 5.33 . The initial yield surface is also shown as a reference. One can see that the two models predict similar plastic flow directions for each of the three different strain paths. It is also worth comparing these results with the ones obtained with the rate-dependent models, see Fig. 3. Under these conditions of normal biaxial loading, there is little appreciable difference between the GC and LA rate-independent models, and between them and the rate-dependent predictions.

To appreciate the implications of the slip systems selected by each model, it can be shown that a simple approximation to the exact internal dissipation throughout the deformation, given by the last two terms on the right hand-side of Eq. (19), can be obtained by adding all the slip rates at any given time (Besson et al., 2001). Fig. 8 gives the variation of the $\sum_x |\dot{\gamma}^x|$, the sum extending over all 12 systems, versus the X_1 stress component for the different imposed E_{11}/E_{22} ratios shown in Fig. 7. Assuming that $\sum_x |\dot{\gamma}^x|$ is an acceptable measure of the internal work, then both models give almost identical internal dissipations except for small differences at the latest stages of the deformation. As discussed earlier, the minimisation of the internal work or internal dissipation is equivalent to a maximisation of the plastic work, thus the predicted internal dissipations should be interpreted within that context. It is also worth pointing out that the jumps shown in Fig. 8 at approximately $T_{11} = 270$ and 285 MPa correspond to secondary slip systems being activated. Similarly, the instabilities which can be seen at around 325 and 245 MPa for the GC model and the $E_{11}/E_{22} = 5.33$ case is due to the hesitation of the local implicit algorithm between

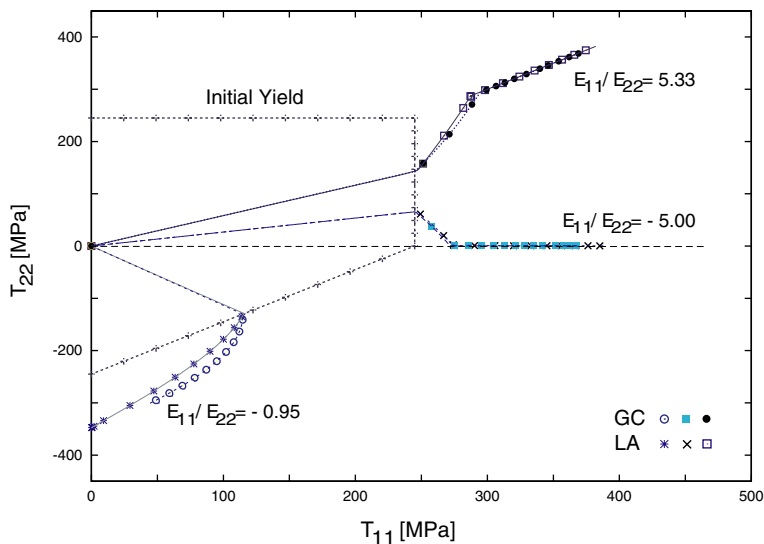


Fig. 7. T_{11} – T_{22} biaxial loading paths under different imposed E_{11}/E_{12} ratios (X_1 and X_2 parallel to $[1\ 0\ 0]$ and $[0\ 1\ 0]$ crystallographic axes, respectively): comparison between the *rate-independent* LA and GC models with Taylor latent hardening and non-directional hardening only ($\dot{R}^x, \dot{S}^x \neq 0, \dot{B}^x = 0$).

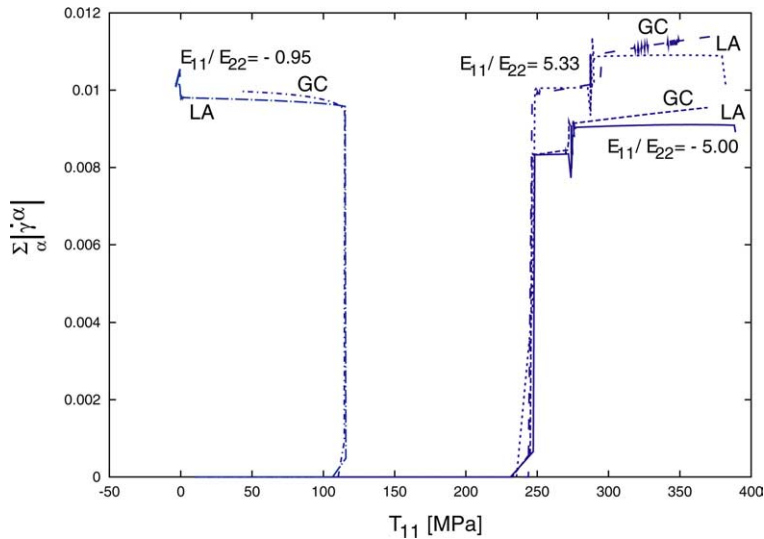


Fig. 8. Sum of all *rate-independent* slip rates versus T_{11} for the octahedral systems under the biaxial T_{11} – T_{22} loading paths and the different imposed E_{11}/E_{22} ratios shown in Fig. 7.

two possible solutions. This behaviour was observed in both the GC and LA models. It is also instructive to replot the sum of the slip rates as a function of time instead of stress for the cases shown in Fig. 7. This is shown in Fig. 9, which also clearly reveals again the instabilities associated with the selection of the active systems by the local numerical algorithm.

Table 4 identifies each of the twelve octahedral slip systems by a number, and Table 5 gives the lists of systems selected by the GC and LA models for the three different biaxial stress responses shown in Fig. 7. It can be seen that exactly the same primary and secondary slip systems were chosen by each model, except for

Table 4
List of octahedral slip systems

System number	Slip plane			Slip direction		
1	1	1	1	$\bar{1}$	0	1
2	1	1	1	0	$\bar{1}$	1
3	1	1	1	$\bar{1}$	1	0
4	1	$\bar{1}$	1	$\bar{1}$	0	1
5	1	$\bar{1}$	1	0	1	1
6	1	$\bar{1}$	1	1	1	0
7	$\bar{1}$	1	1	0	$\bar{1}$	1
8	$\bar{1}$	1	1	1	1	0
9	$\bar{1}$	1	1	1	0	1
10	1	1	$\bar{1}$	$\bar{1}$	1	0
11	1	1	$\bar{1}$	1	0	1
12	1	1	$\bar{1}$	0	1	1

Table 5
Primary and secondary slip systems activated by the GC and LA rate-independent models

Strain ratio (E_{22}/E_{11})	Rate-independent model	Primary systems				Secondary systems			
5.33	LA	1	4	9	11	2	5	7	12
	GC	1	4	9	11	2	5	7	12
−5.00	LA	1	4	9	11	3	6	8	10
	GC	1	4	9	11	3	6	8	10
−0.95	LA	3	6	8	10	2	5	7	12
	GC	3	6	8	10	–			

the $E_{11}/E_{22} = -0.95$ case, whereas the GC model does not predict any secondary slip. Prediction of secondary slip by the LA model for this case is due to the fact that the load point reaches $T_{11} = 0$. Finally, Figs. 10(a) to (c) show the accumulated slip against time for all the octahedral systems under the biaxial T_{11} – T_{22} loading paths and the different E_{11}/E_{12} ratios. By correlating each system from the numbers given in Table 4, with the results given in Fig. 10, it can be concluded that very similar deformation histories are predicted by the GC and LA models, as expected from the results discussed in this section.

Finally, the strong effect of latent hardening on slip system activation can be shown in the predictions of Fig. 11, which shows the predicted biaxial normal-shear stress paths due to imposed E_{11}/E_{12} ratios of 0.02 and 0.40. The initial yield surface and that of the relevant latent system are also shown. As in the rate-dependent case, one can see the contrasting predictions obtained from the two models. It should

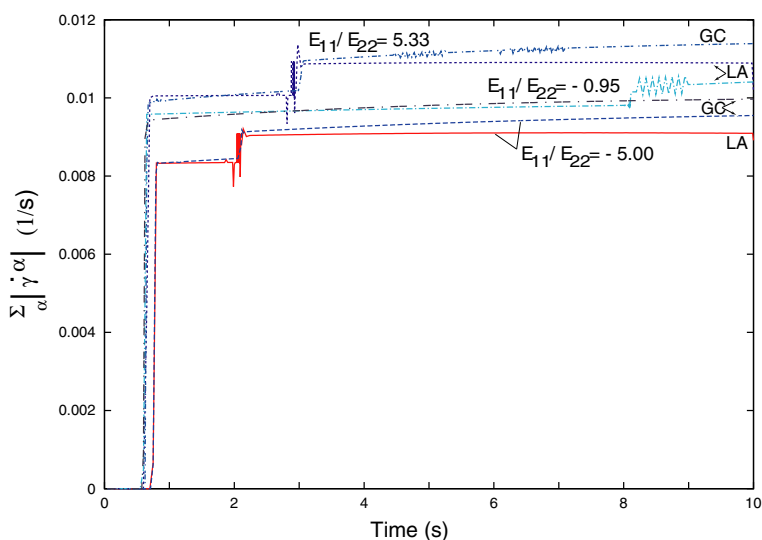


Fig. 9. Sum of all rate-independent slip rates versus time for the octahedral systems under the biaxial T_{11} – T_{22} loading paths and different imposed E_{11}/E_{12} ratios shown in Fig. 7.

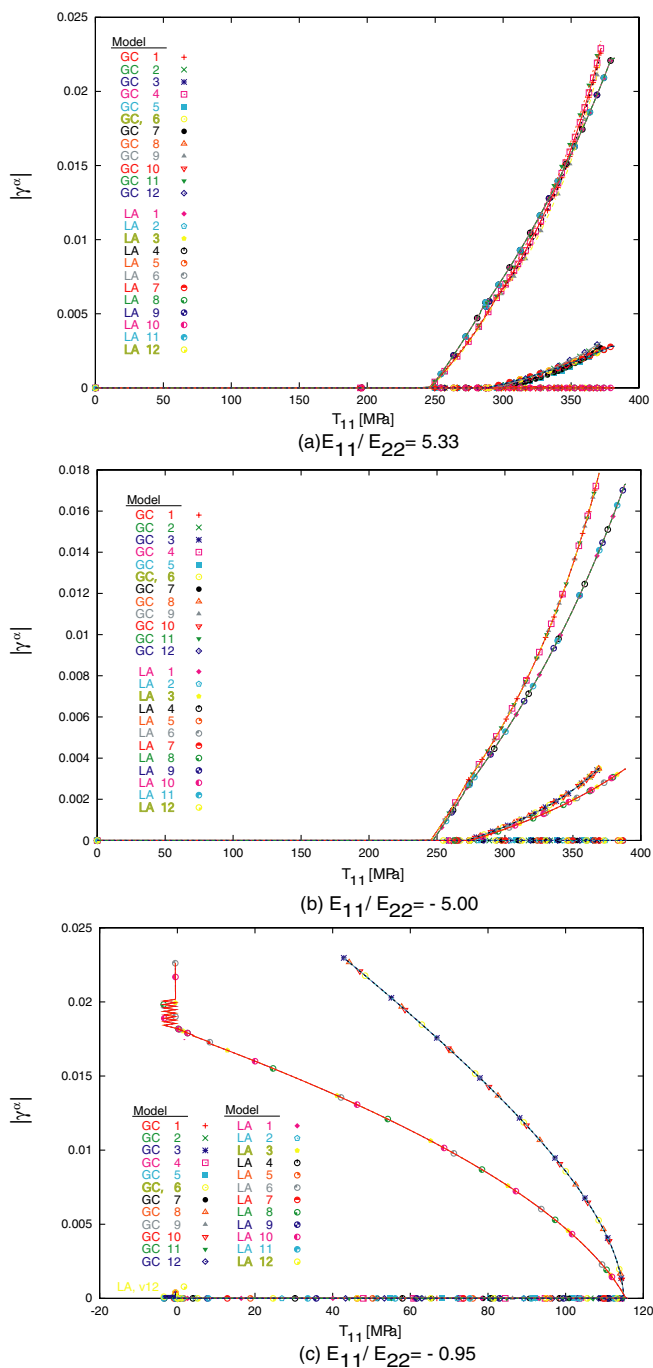


Fig. 10. Accumulated slip vs. time histories for the octahedral systems under the biaxial T_{11} – T_{22} loading paths for E_{11}/E_{12} ratios of (a) 5.33, (b) -0.95 and (c) -5.00 .

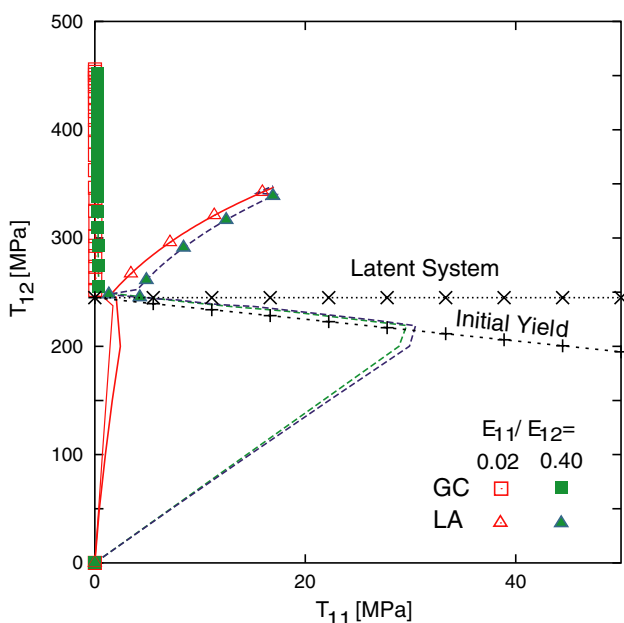


Fig. 11. Comparison between the *rate-independent* model predictions of T_{11} – T_{12} biaxial stress paths under imposed E_{11}/E_{12} strain ratios of 0.02 (open symbols) and 0.40 (filled symbols) with Taylor latent hardening.

however be pointed out that the horizontal and vertical stress scales are vastly different, thus the discrepancies in the shear stresses are relatively smaller than what it would first appear. In Fig. 11, the LA seems to be more sensitive to the corners of the yield surface than the GC model.

5. Conclusions

This work has shown that multiaxial loading and slip system interaction play a crucial role in the selection of the active slip systems and hence the predicted behaviour. It was also found that estimates of the internal dissipated energy and slip systems activated under complex multiaxial loading conditions for two typical rate-independent models were similar. From the results discussed in this work, it can be concluded that overall the behaviour predicted by both rate-dependent and rate-independent formulations in the quasi-rate-independent regime depends on both the particular form of the single crystal formulations and the Newton–Raphson type algorithms used in their numerical integration. However, further work is needed to ascertain the relative effect of the constitutive formulation and the numerical algorithms used to solve the implicit systems of non-linear equations at the local or material point level. The relative accuracy of the single crystal models under

multiaxial loading conditions can only be determined by comparing their predicted behaviour with experimental data obtained under similar conditions.

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